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Internship Domain: Ethical Hacking
Internship Duration: 8 Weeks (June - August 2026)

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1. Introduction

This report summarizes my 8-week Ethical Hacking internship with Edu Skills. The program blended theoretical knowledge with hands-on experience in cybersecurity, covering foundational concepts to advanced penetration testing techniques.

To complement the internship, I completed the **Google Cybersecurity Professional Certificate** on Coursera, which provided a strong theoretical foundation in industry frameworks like **NIST CSF**, **CISSP domains**, **OWASP**, and **MITRE ATT&CK**.

My primary deliverable was the **Enterprise Security Operations Center (SOC) Simulator** – a Python-based security analytics platform demonstrating log collection, threat detection, incident response, and executive reporting.

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2. Internship Structure

Week	Module	Key Learnings
1	Introduction to Ethical Hacking	CIA Triad, purpose and legality of ethical hacking
2	Fundamentals of Security	NIST CSF, NIST RMF, ISO 27001, threat types
3	Social Engineering	Phishing, pretexting, impersonation attacks
4	Malware and Their Types	Ransomware, Trojans, worms, viruses, spyware
5	Networking Essentials	TCP/IP, DNS, HTTP/HTTPS, SSH, FTP, subnetting (CIDR)
6	Introduction to Kali Linux	Metasploit, Burp Suite, John the Ripper, Wireshark
7	Web Application & Vulnerability Assessment	OWASP Top 10, SQL Injection, XSS
8	Penetration Testing	Reconnaissance, exploitation, reporting

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3. Primary Project: Enterprise SOC Simulator

3.1 Project Overview

The **Enterprise SOC Simulator** is a Python-based platform that simulates core security operations, from log generation to incident response and visualization.

3.2 Architecture

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Synthetic Data → SQLite DB (10 Tables) → Detection Engine (5 Rules) → Alerts → Incident Response (4 Playbooks) → Dashboard (9 Charts) → Executive Report
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3.3 Technical Implementation

Database Schema: 10 tables (Assets, Users, Auth Logs, Network Traffic, Firewall Logs, Malware Events, Vulnerabilities, Alerts, Incidents, Playbooks)

Data Generated: 18,605+ synthetic events

- Auth Logs: 12,355 (85% success, 12% failure, 3% locked)
- Network Traffic: 5,000 (5% malicious)
- Malware Events: 250 across 7 types
- Vulnerabilities: 100 CVEs mapped to OWASP
- Firewall Logs: 1,000 events

Threat Detection (5 Rules with MITRE Mapping):

Rule	Detection Logic	MITRE Technique
Brute Force	5+ failures in 2 minutes	T1110
Impossible Travel	Multiple countries in 1 hour	T1078
Port Scanning	20+ ports in 5 minutes	T1046
Malware Activity	Critical/High severity	T1203
SQL Injection	Suspicious HTTP patterns	T1190

Results: 71 alerts generated with proper severity and MITRE mapping.

Incident Response: 4 playbooks defined, 5 incidents processed through full lifecycle (Detection → Analysis → Containment → Eradication → Recovery).

Dashboard: 9 visualizations covering authentication, threats, vulnerabilities, MFA, malware, incidents, assets, and NIST CSF assessment.

Executive Reporting: Automated report with metrics, NIST CSF scores, and prioritized recommendations.

3.4 Framework Alignment

Framework	Implementation
NIST CSF v2.0	All 6 functions
MITRE ATT&CK	5 tactics mapped
OWASP Top 10	5 categories covered
CISSP Domains	All 8 domains represented

3.5 Unit Testing

8 unit tests implemented and passing: database creation, data population, detection logic, configuration, metrics, incident-playbook linkage, and MITRE mapping.

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4. Additional Labs

Qwiklabs: Linux commands, SQL queries, network traffic analysis, SIEM dashboards, incident response playbooks.

Kali Linux Labs: Password cracking (John the Ripper, Hashcat), wireless testing (Aircrack-ng), web app testing (Burp Suite, OWASP ZAP), forensics (Autopsy, Wireshark), vulnerability scanning (OpenVAS, Nessus).

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5. Integration of Google Cybersecurity Certificate

Completed 4 courses covering:

Theory: NIST CSF v2.0, NIST RMF, ISO 27001, CISSP 8 domains, SIEM (Splunk, Chronicle), SOAR, OWASP Top 10.

Skills Applied:

- Linux: grep, tail, cat, find, chmod, chown, useradd, sudo
- SQL: SELECT, FROM, WHERE, JOIN, COUNT, AVG, SUM
- Network: TCP/IP, OSI model, subnetting, protocol analysis

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6. Skills Acquired

Technical Skills	Tools
Vulnerability Assessment	Burp Suite, OWASP ZAP, Nikto, Sqlmap
Penetration Testing	Metasploit, Nmap, John the Ripper
Network Security	Wireshark, Tcpdump, Aircrack-ng
Linux Administration	Kali Linux, Ubuntu, Bash
SQL Database Management	MySQL, SQLite, MariaDB
Incident Response	Splunk, Chronicle
Digital Forensics	Autopsy, Foremost

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7. Conclusion

This internship successfully bridged the gap between theory and practice. The **Enterprise SOC Simulator** project demonstrates my ability to build a functional security analytics platform covering the entire security operations workflow. Combined with the Google Cybersecurity Certificate, this experience has prepared me for a career in cybersecurity.

Verification Links

EduSkills Internship Record:
https://academyapi.eduskills.academy/offer-letters/student-details/1850

Google Cybersecurity Certificate Links:
https://www.coursera.org/account/accomplishments/verify/6U2MJU04BFJ8
https://www.coursera.org/account/accomplishments/verify/Q5JE0VGBVCHY
https://www.coursera.org/account/accomplishments/verify/VPSAIWKT3SOG
https://www.coursera.org/account/accomplishments/verify/DE83LSX8ZCGL

SOC Simulator Project:
https://www.kaggle.com/code/mrrogueknight/security-operations-dashboard
https://github.com/MrRogueKnight/security-operations-dashboard